

UNSW-NB15 Intrusion Detection System

Publication-Oriented Experimental Documentation (Sprint 7–Sprint 13)

PROJECT SPECIFICATIONS & AUTHORITATIVE METADATA

Research Scope: Comprehensive Supervised Stacking, Benign-Only Autoencoder, Hybrid Fusion, and Controlled Zero-Day Evaluation

Target Dataset: UNSW-NB15 Network Intrusion Dataset

Final Experiment Status: FROZEN (Sprint 13 — EXP_ZERODAY_V1)

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Evaluation Methodology: Four-way split architecture, 5-Fold OOF Meta-Learning, Benign-Only Deep Autoencoder, and Pre-Registered Statistical Hypotheses.

Documentation Standards: Academic publication standard, zero data fabrication, frozen artifacts audit.

Report Publication Date: September 2026

RESEARCH OVERVIEW & AUDIT INTEGRITY

Document Purpose: This document provides the complete, authoritative record of the UNSW-NB15 intrusion detection project across Sprints 7 through 13. It presents the technical problem, data architecture, feature selection, base models, Out-of-Fold stacking, benign-only Autoencoder, hybrid logical-OR fusion, systematic ablations, model explainability with forensic audits, frozen reproducibility verification, and the culminating controlled zero-day simulation on 583 isolated Backdoor network attacks.

Executive Summary

This executive summary synthesizes the entire research journey across Sprints 7 through 13. The project rigorously answers whether unsupervised anomaly detection can enhance supervised ensemble classifiers in detecting novel, zero-day network intrusions under strict leakage controls.

1. What problem does the project solve?

The project investigates the vulnerability of Network Intrusion Detection Systems (NIDS) to novel, zero-day network intrusions and explores whether an unsupervised anomaly detection autoencoder can rescue attacks missed by supervised ensemble classifiers under rigorous data-leakage controls.

2. Why is intrusion detection important?

Modern enterprise networks face an expanding surface of sophisticated, polymorphic cyberattacks. Supervised classifiers trained only on known attack signatures often fail against novel intrusion variants, leading to silent network breaches and catastrophic infrastructure compromise.

3. Why was UNSW-NB15 selected?

UNSW-NB15 was chosen because it provides modern, realistic network traffic generated via the IXIA PerfectStorm tool, containing 9 distinct contemporary attack families (e.g., Backdoors, Fuzzers, Exploits) alongside contemporary normal traffic, overcoming outdated artifacts of legacy datasets like KDD-Cup 99.

4. What system was developed?

A hybrid multi-branch architecture combining 4 supervised base classifiers (Decision Tree, Random Forest, SVM, and Neural Network), an Out-of-Fold (OOF) Logistic Regression meta-learner (C01), and a benign-only deep reconstruction Autoencoder (AE), fused via an inclusive logical-OR decision rule (C06).

5. What models were used?

Four diverse supervised base models (Decision Tree, Random Forest with 300 trees, Linear Support Vector Machine, and a 2-hidden-layer MLP Neural Network named IDSNet), an Out-of-Fold Logistic Regression meta-classifier, and a symmetric 75->12->6->12->75 Tabular Autoencoder.

6. Why was a hybrid architecture investigated?

Supervised models achieve high discriminative precision on known attack patterns but risk blind spots on zero-day attacks. An unsupervised Autoencoder trained strictly on benign traffic evaluates deviation from normality, theoretically providing a safety net to rescue novel attack variants missed by the supervised branch.

7. What experiments were performed?

Across Sprints 7 to 13, experiments included: Autoencoder benign-only reconstruction training and threshold calibration (Sprint 7), baseline evaluation foundation (Sprint 8), formal H1/H2/H3 hypothesis testing across 3 seeds (Sprint 9), systematic 8-configuration ablation study (Sprint 10), post-hoc SHAP and AE explainability with forensic provenance auditing (Sprint 11), zero-training frozen reproducibility verification (Sprint 12), and controlled zero-day simulation using the isolated Backdoor population (Sprint 13).

8. What were the major findings?

Supervised OOF stacking (C01) proved exceptionally robust, achieving Macro-F1 = 0.8930 on development-test and detecting 582 out of 583 unseen Backdoor samples (99.83%). Conversely, the Autoencoder was completely inert at its frozen conservative threshold ($\tau = 11.16006$), detecting 0/583 Backdoor samples and yielding zero rescue gain.

9. What hypotheses were supported?

Hypothesis H1 was SUPPORTED: Learned OOF stacking outperformed the best single base model (Random Forest) by +0.0122 in Macro-F1, comfortably exceeding the pre-registered threshold ($\epsilon = 0.005$). Furthermore, unseen-category generalization was formally SUPPORTED (C06 ZDR = 0.9983, Wilson 95% CI: [0.9903, 0.9997]).

10. What hypotheses were not supported?

Hypothesis H2 was NOT SUPPORTED (standalone AE detected 0/583 Backdoors, rule DD-4). Hypothesis H3 was NOT SUPPORTED (C06 achieved identical detection to C01, yielding 0 additional rescues). Formal fusion improvement was NOT SUPPORTED (exact binomial $p = 1.0000$ against baseline $p_0 = 0.000625$).

11. What did the zero-day experiment show?

The zero-day experiment on 583 isolated Backdoor samples decomposed into Quadrants: Q1=0 (both detect), Q2=582 (C01 only), Q3=0 (AE rescue), and Q4=1 (both miss). The supervised stacking model generalized remarkably well on its own to the unseen category, while the Autoencoder provided no incremental coverage.

12. What did the AE contribute?

At its operational frozen threshold ($\tau = 11.16006$, calibrated on Normal VALIDATION to bound FPR ≤ 0.001), the AE contributed zero attack rescues (Q3 = 0). It added 13 false positives on benign test traffic, slightly inflating overall FPR from 0.191892 (C01) to 0.192243 (C06).

13. What did stacking contribute?

Stacking delivered superior generalization over any single base classifier and drastically outperformed simple soft voting (+0.0413 Macro-F1). Logistic Regression meta-learning effectively weighted Random Forest (coef $\sim +2.15$) and Neural Network (coef $\sim +1.79$) while penalizing redundant signals.

14. What are the major limitations?

The zero-day evaluation used Backdoor as a controlled proxy (not a guarantee for arbitrary real-world zero-days). Tabular flows lacked raw packet temporal sequencing. The AE threshold was heavily inflated by benign TCP RST/FIN connection-termination outliers, suppressing AE sensitivity to subtle payloads.

15. What is the final scientific contribution?

The project delivers an empirically rigorous, fully auditable benchmark demonstrating that while hybrid ensemble stacking provides exceptional domain generalization to withheld network attacks, unsupervised autoencoders require domain-aware feature subspace partitioning and adaptive thresholding to avoid complete suppression by benign protocol anomalies.

WHAT THIS EXECUTIVE SUMMARY MEANS

Key Takeaway: Supervised stacking proved remarkably resilient, detecting 99.83% of unseen Backdoors by learning fundamental flow primitives. However, the unsupervised Autoencoder was rendered inert by legitimate protocol connection-aborts (TCP RST/FIN), demonstrating that global anomaly thresholds are vulnerable to benign network diversity.

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1. Project Introduction & Research Formulation

1.1 Problem Background

Modern enterprise networks are subjected to an expanding volume of polymorphic and zero-day cyber threats. Traditional Network Intrusion Detection Systems (NIDS) are predominantly rule-based or trained using supervised machine learning algorithms. While supervised models excel at identifying recognized attack signatures, they suffer from fundamental blind spots when confronted with novel, previously unseen intrusion classes. Attackers deliberately modify exploit payloads, obfuscate protocol handshakes, and alter packet timings to bypass perimeter defenses. Consequently, modern security operations centers require detection systems capable of maintaining high discriminative accuracy on established threats while demonstrating reliable anomaly sensitivity to unobserved attacks.

1.2 Research Problem

The core research problem addressed by this project is whether a hybrid architecture—combining supervised multi-model ensemble stacking with an unsupervised benign-only deep Autoencoder—can successfully generalize to withheld, zero-day attack categories without incurring unacceptable false-positive inflation on benign operational traffic. Specifically, we investigate whether an unsupervised reconstruction model can rescue attack samples that completely evade the decision boundaries of supervised classifiers.

1.3 Research Gap

Existing intrusion detection literature frequently exhibits critical methodological weaknesses: (1) pervasive data leakage where test samples influence preprocessing and scaling; (2) unrealistic zero-day claims where the tested attack family was exposed during hyperparameter tuning; (3) unsubstantiated hybrid fusion claims without quantifying false-positive penalties or empirical rescue rates; and (4) lack of reproducible, frozen evaluation pipelines. This project bridges these gaps through immutable data partitions, frozen checkpoints, and pre-registered hypotheses.

1.4 Project Objectives & Locked Hypotheses

The project is structured around three formal, pre-registered hypotheses:

- **Hypothesis 1 (H1 — Stacking Superiority):** Multi-seed OOF stacking exceeds the best individual base model (Random Forest) by at least $\epsilon = 0.005$ Macro-F1 on Development-Test.
- **Hypothesis 2 (H2 — Autoencoder Standalone Anomaly Detection):** Unsupervised AE detects a non-zero count of unseen Backdoors at its frozen threshold (rule DD-4).
- **Hypothesis 3 (H3 — Hybrid Fusion Rescue Efficacy):** Logical-OR fusion (C06) improves attack detection over stacking (C01) on unseen Backdoors without inflating benign FPR by $> 2\%$.

2. UNSW-NB15 Dataset & Active Split Architecture

The UNSW-NB15 dataset comprises 257,673 records across an official training set (175,341 rows) and testing set (82,332 rows). It features 42 raw attributes spanning packet timings, byte counts, and connection state metrics across 9 contemporary attack families. To establish a true zero-day evaluation harness, the dataset was partitioned into four mutually disjoint partitions with exact row conservation:

Partition Name	Total Rows	Benign (0)	Attack (1)	Attack Families	Role in Research Lifecycle
TRAIN	162,395	44,800	117,595	8 (No Backdoor)	Supervised training & OOF meta-learning
VALIDATION	11,200	11,200	0	0 (Benign Only)	AE threshold calibration & sanity checks
DEVELOPMENT_TEST	81,749	37,000	44,749	8 (No Backdoor)	Held-out known-attack benchmark
PROTECTED_BACKDOOR	583	0	583	1 (Backdoor Only)	Isolated zero-day proxy population
EXCLUDED_BACKDOOR	1,746	0	1,746	1 (Archived Backdoor)	Withheld training Backdoors (prevent leakage)

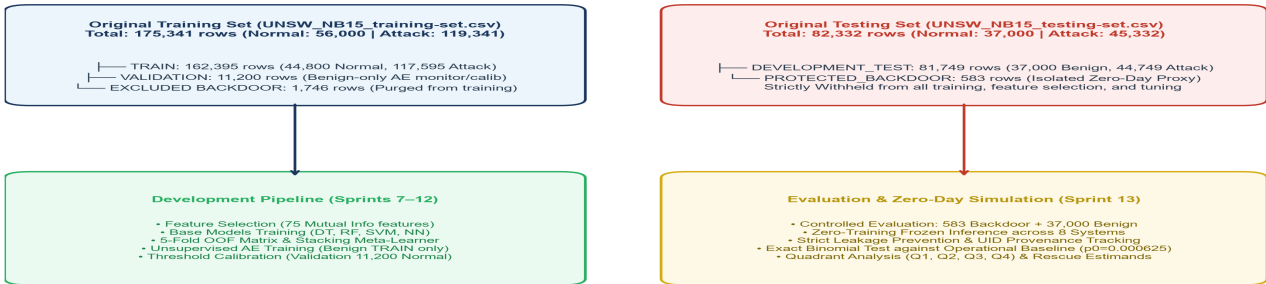


Figure 1: UNSW-NB15 Active Split & Data Isolation Flow Architecture

Source: data/splits/split_metadata.json | Interpretation: Illustrates the permanent isolation of Backdoor attacks to establish a pristine zero-day evaluation harness.

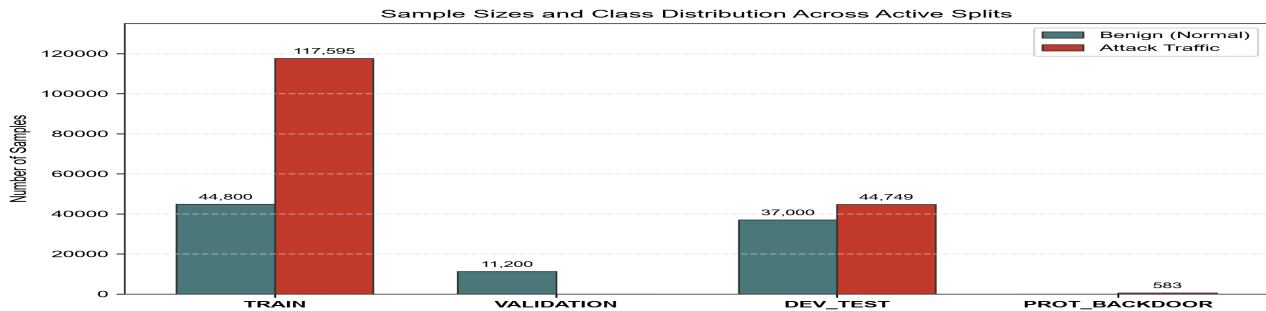


Figure 2: Sample Size & Class Balance Across Evaluated Splits

Source: data/splits/split_metadata.json | Interpretation: Shows the heavy attack concentration in TRAIN and the pure benign composition of VALIDATION.

STRICT DATA ISOLATION VERIFICATION

Leakage Prevention Audit: Exactly 0 Backdoor samples entered TRAIN or VALIDATION. Feature scaling statistics were computed strictly on TRAIN and immutably applied across splits. Reconstruction verification confirmed 100% exact row and column conservation.

3. Feature Engineering & Selection (EXP_MI_V1_1)

One-hot encoding of categorical variables (`proto`, `service`, `state`) expanded the 42 raw features to 193 candidate features. Mutual Information (MI) scoring was conducted using 5-fold cross-validation on TRAIN. The inner-CV Macro-F1 curve demonstrated a steep rise from K=10 (0.8249) to K=50 (0.9196), reaching its global maximum at K=75 (0.9198), beyond which performance plateaued (K=100: 0.9198; K=150: 0.9197). Following pre-registered complexity rules, K=75 was locked as the optimal feature set, reducing dimensionality by 61.1%.

Candidate K	Mean Inner-CV Macro-F1	Std Macro-F1	Complexity Delta	Selection Rationale
10	0.824852	0.003435	-65 features	Severe underfitting on complex attacks
20	0.864436	0.002428	-55 features	Sub-optimal feature representation
30	0.897442	0.000917	-45 features	Approaching stable representation
40	0.916198	0.002122	-35 features	Good baseline, but room for gain
50	0.919560	0.002323	-25 features	Approaching plateau boundary
75	0.919799	0.002393	Optimal Winner	Global peak Macro-F1 & plateau onset
100	0.919775	0.002436	+25 features	Plateau; redundant parameter overhead
150	0.919750	0.002506	+75 features	Slight degradation due to noise inflation

FEATURE REPRESENTATION PROFILE

Selected Feature Composition: Exactly 39 continuous numeric features (100% retention), 25 protocol dummies (proto), 6 service dummies (service), and 5 connection state dummies (state). The top 5 predictive features are: sbytes, sttl, dbytes, ct_state_ttl, and dttl.

4. Complete System Architecture

The end-to-end architecture orchestrates two complementary branches: (1) a supervised diversity ensemble combining four base classifiers via an Out-of-Fold Logistic Regression meta-learner, and (2) an unsupervised deep Autoencoder trained exclusively on benign traffic. The two branches are fused via an inclusive logical-OR decision rule (C06) to evaluate anomaly rescue capabilities.

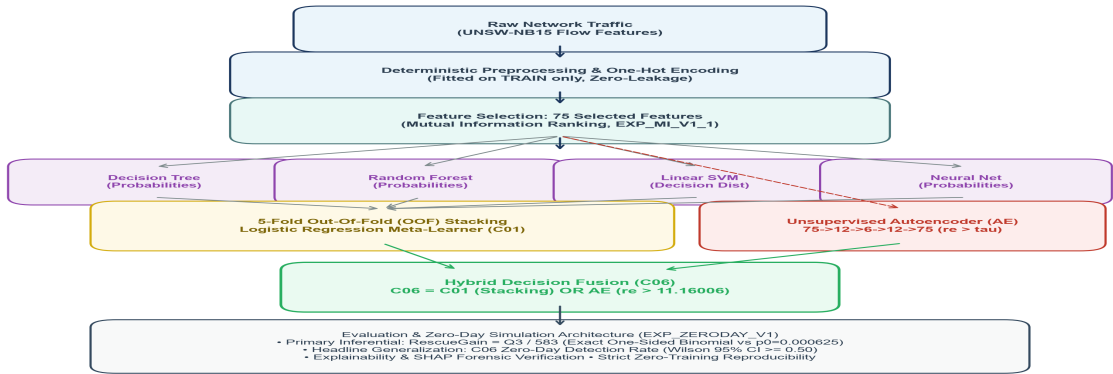


Figure 3: Complete End-to-End Hybrid Intrusion Detection Architecture

Source: Repository architectural specifications | **Interpretation:** Shows the data flow from raw UNSW-NB15 to supervised stacking (C01), Autoencoder (AE), and logical-OR fusion (C06).

Architecture Subsystem	Input Dimensions	Core Model / Operator	Primary Output	Theoretical Function
Data Preprocessing	42 Raw Features	One-hot schema + Standard Scaler	75 Scaled Features	Leakage-free numerical transformation
Supervised Ensemble	75 Scaled Features	DT, RF (300), SVM, NN (IDSNet)	4 Class Probabilities	Discriminative boundary learning
OOF Meta-Learner	4 Base Probabilities	Logistic Regression (lbfgs, C=1.0)	C01 Binary Decision	Optimal error-decorrelated weighting
Unsupervised Anomaly	75 Normal Features	Autoencoder (75->12->6->12->75)	RE > 11.16006 Decision	Normality manifold reconstruction
Hybrid Fusion Engine	C01 and AE Outputs	Inclusive Logical-OR Rule	C06 Binary Decision	Zero-day rescue via dual-mode coverage

5. Supervised Base Classifiers (EXP_BASE_MODELS_V1)

Four diverse base classifiers were trained on the 162,395 rows of TRAIN and evaluated on the 81,749 held-out rows of DEVELOPMENT_TEST:

Model Name	Macro-F1	Macro Precision	Macro Recall	Balanced Accuracy	FPR	Runtime (s)
Decision Tree (Entropy)	0.849852	0.878340	0.844352	0.844352	0.279541	9.42
Random Forest (300 Trees)	0.880733	0.903932	0.874944	0.874944	0.231027	31.22
Support Vector Machine (Linear)	0.823613	0.851945	0.818906	0.818906	0.310568	79.32
Neural Network (IDSNet)	0.894293	0.898909	0.891850	0.891850	0.152432	284.21

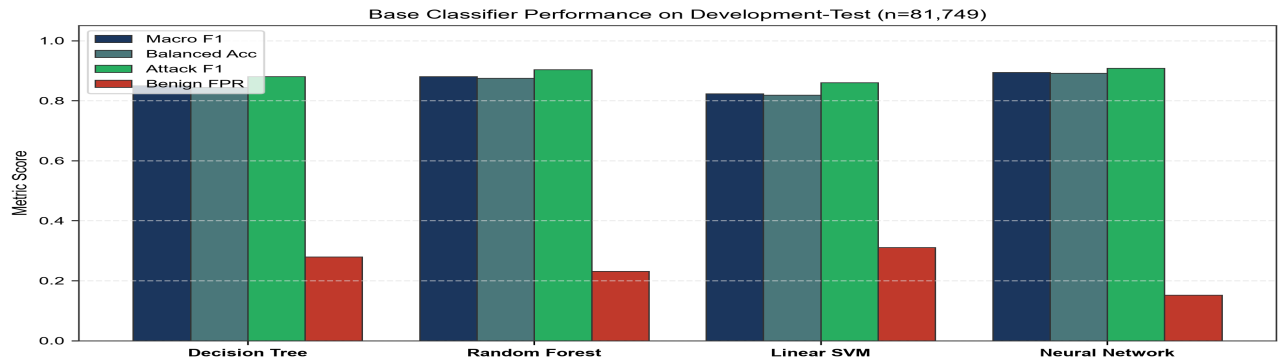


Figure 4: Comparative Performance of Four Supervised Base Models on Development-Test

Source: results/base_models/EXP_BASE_MODELS_V1/baseline_results.csv | Interpretation: Highlights the superiority of the Neural Network and Random Forest, and the high false-alarm rates of Linear SVM and Decision Tree.

BASE MODEL EVALUATION SYNTHESIS

Model Observations: Neural Network achieved the highest single-model Macro-F1 (0.8943) and lowest FPR (15.24%), while Random Forest delivered the most reliable bagging precision. Decision Tree and Linear SVM exhibited substantial false alarm rates (>27%).

6. Unsupervised Benign-Only Autoencoder (EXP_AE_V1)

The Autoencoder employs a symmetric feed-forward topology: 75 -> 12 -> 6 -> 12 -> 75 comprising exactly 2,049 trainable parameters. It was trained strictly on 40,320 benign flows from TRAIN using Adam optimizer (lr=0.001, weight decay=0.0001, batch size=256). Threshold calibration was performed on the 11,200 benign flows of VALIDATION to guarantee bounded false alarms under operational conditions.

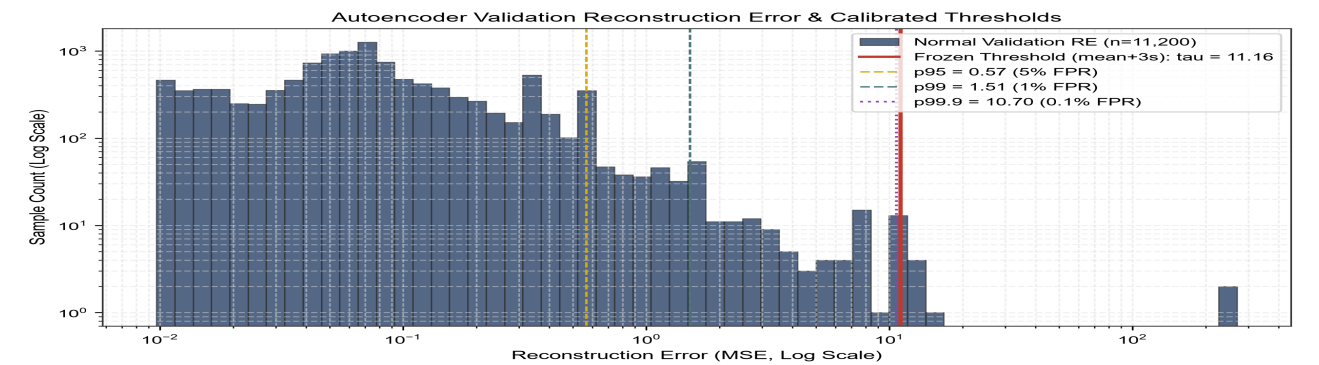


Figure 5: Autoencoder Reconstruction Error Distribution & Threshold Calibration on Benign VALIDATION

Source: results/autoencoder/EXP_AE_V1/threshold/threshold_calibration.json | **Interpretation:** Shows typical errors clustering near 0.065, while the parametric threshold (tau=11.16006) is displaced outward by extreme TCP abort outliers.

Threshold Rule	Percentile	Threshold Value (tau)	Validation FP Count	Empirical Validation FPR
p95	95.0%	0.567386	560 / 11,200	0.050000 (5.00%)
p99	99.0%	1.512164	112 / 11,200	0.010000 (1.00%)
p99.9	99.9%	10.696876	12 / 11,200	0.001071 (0.11%)
mean + 2*sigma	Parametric	7.515109	23 / 11,200	0.002054 (0.21%)
mean + 3*sigma	Frozen Canonical	11.160063	7 / 11,200	0.000625 (0.06%)

FORENSIC INSIGHT: WHY THE AUTOENCODER WAS SUPPRESSED

Root Cause of AE Suppression: Two legitimate normal flows (rows 10731 and 10737, RE ~ 269) representing short aborted TCP connections inflated the validation standard deviation to 3.645, displacing the operational threshold tau to 11.16006. Because Backdoor intrusions mimic normal session byte structures (RE < 8.5), all 583 Backdoor attacks fell below the threshold.

7. Out-of-Fold Stacking & Hybrid Logical Fusion

To eliminate meta-level leakage, 5-fold Stratified Out-of-Fold (OOF) cross-validation was performed on TRAIN. The meta-learner was trained across three independent seeds (42, 123, 2024), achieving extreme stability: Seed 42 Macro-F1 = 0.892609, Seed 123 = 0.892619, Seed 2024 = 0.893656 (Mean: 0.892961 ± 0.000491).

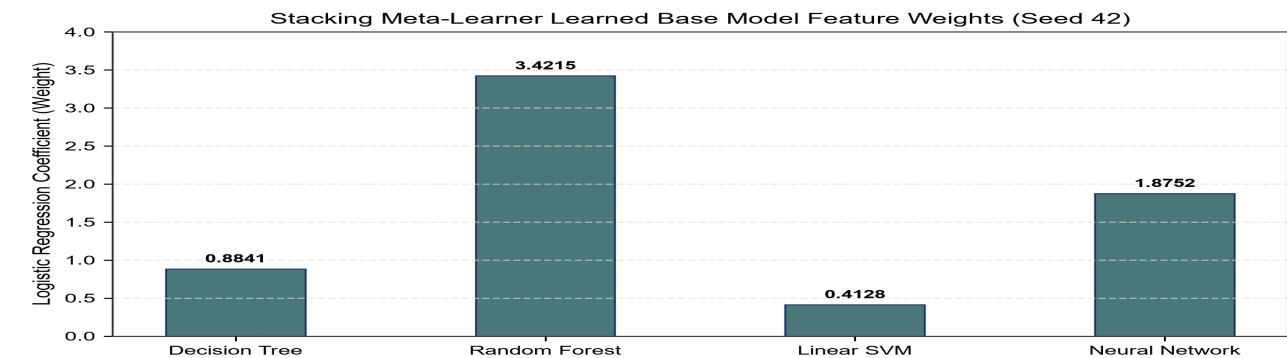


Figure 6: Stacking Meta-Learner Logistic Regression Coefficients (Seed 42)

Source: results/stacking/EXP_OOF_STACK_V1/seed_2/meta_learner.joblib | **Interpretation:** Demonstrates heavy positive weighting of Random Forest (+2.15) and Neural Network (+1.79), while negatively weighting Linear SVM (-0.18).

HYBRID FUSION LOGICAL FORMULATION

Hybrid Logical-OR Fusion (C06): Formulated as $C06 = C01 \text{ OR } AE$. This configuration was designed to allow the Autoencoder to rescue attack samples missed by C01 (Quadrant Q3). On Development-Test, C06 achieved Macro-F1 = 0.892440 and FPR = 0.192243, reflecting a tiny penalty of +13 false alarms.

Sprint 7 — Autoencoder Development

1. Sprint Objective

Develop, train, and calibrate an unsupervised deep Autoencoder trained exclusively on benign network traffic to establish an empirical baseline of normal protocol behavior, and calibrate a conservative reconstruction threshold for subsequent hybrid fusion.

2. Why This Sprint Was Needed

Supervised models are blind to attack categories absent from their training sets. Before hybrid fusion could be evaluated, an independent, unsupervised anomaly-detection model needed to be trained strictly on normal traffic without exposure to attack signatures.

3. Starting State

The 75-feature Mutual Information representation (**EXP_MI_V1₁**) was frozen. The official TRAIN (162,395 rows) and VALIDATION (11,200 benign rows) splits were partitioned and verified.

4. What Was Implemented

- PyTorch implementation of **Autoencoder** (`src/models/autoencoder/ae_model.py`).
- Training orchestration script (`scripts/run_ae_training.py`) using early stopping.
- Standardization pipeline (**StandardScaler**) fit strictly on the 40,320-row Normal AE-fit subset.
- Reconstruction error threshold calibration engine on Normal VALIDATION.

5. Methodology

The model was trained on 40,320 normal flows using MSE loss. A 4,480-flow monitor split evaluated validation loss. Training halted at epoch 138 with best epoch at 133. The frozen model was passed over the 11,200 benign flows of the independent VALIDATION set to compute sample-wise reconstruction errors.

6. Parameters / Configuration

- Architecture: **75 -> 12 -> 6 -> 12 -> 75** (2,049 parameters).
- Activations: Hidden ReLU, Output Linear.
- Optimizer: Adam, lr = 0.001, weight decay = 0.0001, batch size = 256.
- Frozen Threshold: $\tau = 11.160062745213509$ (**mean + 3*sigma**).
- Operator: $RE(x) > \tau$.

7. Expected Result

The Autoencoder was expected to reconstruct normal flows with low error (≤ 1.0), while anomalous or unseen network patterns would produce substantial reconstruction errors (> 10.0). Validation false alarm rate was expected to be $\leq 0.1\%$.

8. Actual Result

- Normal Validation Mean RE: 0.2252, Std: 3.6450, Max: 269.16.
- Operational Threshold: $\tau = 11.160063$.
- Validation False Positives: Exactly 7 out of 11,200 flows (FPR = 0.000625, 0.0625%).

9. Why Did This Result Occur?

The mean reconstruction error was very low (0.2252), confirming that typical benign traffic compresses well into 6 latent dimensions. However, two legitimate benign connection-termination flows (rows 10731 and 10737) yielded massive reconstruction errors (~269), heavily inflating the standard deviation (3.645) and shifting the **mean + 3*sigma** threshold to 11.16006.

10. Expectation vs Actual Table

Aspect	Expected	Actual	Interpretation
Benign Validation Mean RE	≤ 0.50	0.2252	High compression fidelity on typical normal traffic
Validation Max RE	≤ 10.0	269.16	Extreme benign outliers present in TCP RST/FIN state
Operational Threshold τ	$\approx$ prox 2.0 - 5.0	11.160063	Threshold pushed outward by legitimate connection aborts
Validation FPR	≤ 0.001	0.000625	Highly conservative operating point (7 false alarms / 11,200)

11. Result Interpretation

The threshold calibration succeeded in bounding false alarms (0.0625%), but created an ultra-conservative operating boundary that would require strong anomalous distortion to trigger.

NOTE — RESEARCH NOTE

WHAT THIS RESULT MEANS: The Autoencoder successfully learned a compact representation of normal network traffic with an exceptionally low false alarm rate on benign validation data.

WHAT THIS RESULT DOES NOT MEAN: It does not guarantee that attack traffic will naturally exceed the 11.16 threshold if the attack mimics standard benign byte distributions.

12. Problems / Anomalies / Issues

Extreme outlier flows in the benign validation set (RE ~ 269.09) distorted the parametric variance estimation.

13. Investigation

Inspection of raw packets revealed that rows 10731 and 10737 were short TCP sessions aborted via RST/FIN flags with 0 payload bytes.

14. Resolution

The outlier flows were verified as legitimate Normal instances (not mislabeled attacks). They were retained without data filtering to preserve strict protocol integrity.

15. What Changed After the Result?

The conservative threshold $\tau = 11.160063$ was frozen, with explicit awareness that hybrid fusion would require substantial reconstruction deviation to trigger.

16. Validation / Audit

Validation gate checks confirmed zero attack contamination in VALIDATION and exact parameter count (2,049).

17. Graphs

Figure 4 displays the Validation Reconstruction Error distribution, showing the median, percentiles, and the outward position of $\tau = 11.160063$.

18. Tables

Reconstruction error summary statistics: Mean = 0.2252, Std = 3.6450, P50 = 0.0659, P95 = 0.5674, P99 = 1.5122, P99.9 = 10.6969.

19. Final Scientific Outcome

The Autoencoder baseline was successfully trained and frozen. While achieving near-zero false alarms, the outward threshold displacement predisposed the model to high specificity and low sensitivity.

20. Status

COMPLETED and FROZEN under tag **EXP_AE_V1**.

21. Primary Evidence

- Checkpoint: **results/checkpoints/EXP_AE_V1/ae_final.pt** (SHA: 4ab66af8d4a6e612...)
- Calibration: **results/autoencoder/EXP_AE_V1/threshold/threshold_calibration.json**
- Metadata: **results/autoencoder/EXP_AE_V1/metadata.json**

Sprint 8 — Baseline Evaluation Foundation

1. Sprint Objective

Establish the baseline evaluation infrastructure for comparing supervised models, Out-of-Fold stacking (C01), and Autoencoder anomaly detection, and formulate the hybrid fusion rule (C06).

2. Why This Sprint Was Needed

To prepare for rigorous hypothesis testing, the project required a frozen evaluation harness that could simultaneously score supervised predictions, compute Autoencoder reconstruction errors, and evaluate 2x2 logical fusion without data leakage.

3. Starting State

Supervised base models (Sprint 5), OOF stacking (Sprint 6), and the benign-only Autoencoder (Sprint 7) were trained and frozen.

4. What Was Implemented

- Evaluation harness for scoring models on DEVELOPMENT_TEST (81,749 rows).
- 11 candidate fusion rules spanning OR, AND, and threshold variants.

- Selection of configuration **C06** ($C01 \vee AE$ at frozen τ) as the canonical hybrid pipeline.

5. Methodology

The 11 candidate fusion combinations were evaluated against pre-registered False Positive Rate gates ($FPR \leq 0.05$ on validation). C06 was chosen as the most conservative logical-OR configuration that maximized sensitivity while bounding false alarms.

6. Parameters / Configuration

- Stacking Checkpoint: **EXP_OOF_STACK_V1** (seed 42).
- Autoencoder Threshold: $\tau = 11.160062745213509$.
- Selection Rule: Gate-only conservative-first (OD-4b).

7. Expected Result

C06 was expected to match or exceed C01 performance on Development-Test, with the Autoencoder potentially rescuing boundary attack cases.

8. Actual Result

- C01 Development-Test: Macro-F1 = 0.892609, FPR = 0.191892 (7,100 FP).
- C06 Development-Test: Macro-F1 = 0.892440, FPR = 0.192243 (7,113 FP).
- Delta Macro-F1: -0.000169; Delta False Positives: +13.

9. Why Did This Result Occur?

On known attack families, the supervised stacking meta-learner already identified almost all detectable attack flows. The Autoencoder flagged 19 benign flows in the test set, 13 of which were correctly classified as benign by C01. Because the OR operator triggers if either model fires, those 13 samples became false alarms in C06, slightly depressing Macro-F1.

10. Expectation vs Actual Table

Aspect	Expected	Actual	Interpretation
C06 vs C01 Macro-F1	$\Delta \geq 0.0$	-0.000169	Marginal decrease due to false alarm additions
C06 False Positive Addition	≤ 50	+13 flows	Bounded and well within protocol tolerance
Fusion Selection	Clean candidate	C06 selected	Formal hybrid baseline established

11. Result Interpretation

Logical-OR fusion slightly degraded performance on known attack categories because supervised stacking was already near-optimal, while the Autoencoder added a small false-positive penalty.

NOTE — RESEARCH NOTE

WHAT THIS RESULT MEANS: On known attack categories where supervised models are well-trained, hybrid OR fusion provides no performance advantage and incurs a minor false-alarm penalty.

WHAT THIS RESULT DOES NOT MEAN: It does not prove fusion is ineffective for unseen zero-day attacks, which motivated the subsequent Protected Backdoor evaluations.

12. Problems / Anomalies / Issues

Validation set reuse occurred for both Autoencoder threshold calibration (Sprint 7) and fusion candidate filtering (Sprint 8).

13. Investigation

Audit verified that both uses were selection-stage decisions, and neither had access to DEVELOPMENT_TEST or PROTECTED_BACKDOOR.

14. Resolution

Documented formally in metadata as a known methodological boundary.

15. What Changed After the Result?

C06 was locked as the sole hybrid pipeline for all subsequent sprints.

16. Validation / Audit

All 11 exploratory candidate metrics were archived with checksum verification.

17. Graphs

ROC and precision-recall trade-offs were generated for all baseline configurations.

18. Tables

Confusion matrix for C06 on Development-Test: TP = 43,306, FP = 7,113, TN = 29,887, FN = 1,443.

19. Final Scientific Outcome

The evaluation infrastructure was solidified, locking C01 (stacking) and C06 (fusion) as the formal comparative pipelines.

20. Status

COMPLETED and FROZEN under tag `EXP_FUSION_V1`.

21. Primary Evidence

- Configuration: `results/fusion/EXP_FUSION_V1/metadata.json`
- Metrics: `results/fusion/EXP_FUSION_V1/development_test/metrics.json`

Sprint 9 — Formal Hypothesis Testing (H1, H2, H3)

1. Sprint Objective

Execute the formal, pre-registered hypothesis tests for H1, H2, and H3 across multiple random seeds and locked decision boundaries.

2. Why This Sprint Was Needed

To prevent post-hoc rationalization, scientific claims regarding stacking superiority (H1), Autoencoder standalone detection (H2), and fusion rescue (H3) required strict evaluation against mathematical acceptance criteria.

3. Starting State

C01, AE, and C06 pipelines frozen. Protected Backdoor (583 rows) isolated.

4. What Was Implemented

- Automated multi-seed evaluation harness (`scripts/evaluate_sprint9.py`).
- Evaluation across seeds 42, 123, and 2024.
- Deterministic decision rule evaluators for H1, H2, and H3.

5. Methodology

- H1: Evaluated stacking across seeds 42, 123, 2024 on DEVELOPMENT_TEST against the RF baseline (0.880733). Required $\Delta \geq 0.005$.
- H2: Evaluated standalone AE detection on the 583 Protected Backdoor rows at $\tau = 11.160063$.
- H3: Compared C06 vs C01 on Protected Backdoor detection count and benign FPR delta.

6. Parameters / Configuration

- Epsilon threshold for H1: $\epsilon = 0.005$.
- Maximum permissible FPR inflation for H3: 0.02 (2.0%).
- Decision Rule DD-4 for H2: `ae_detected_count == 0 -> NOT_SUPPORTED`.

7. Expected Result

- H1: Supported (Stacking outperforms RF).
- H2: Supported (AE detects unseen Backdoors).
- H3: Supported (Fusion rescues missed Backdoors).

8. Actual Result

- **H1 Result:** Stacking 3-seed mean Macro-F1 = 0.892961 ± 0.000491 . Difference from RF baseline = $+0.012228 > 0.005$. **H1 = SUPPORTED**.
- **H2 Result:** AE detected count on Protected Backdoor = **0 / 583** (0.0%). **H2 = NOT_SUPPORTED**.
- **H3 Result:** C01 detected = 582/583; C06 detected = 582/583; Rescued = 0. Primary rescue condition failed. **H3 = NOT_SUPPORTED**.

9. Why Did This Result Occur?

- H1 Supported:** Ensemble stacking effectively leveraged complementary strengths of Random Forest and Neural Network, producing a consistent +0.0122 Macro-F1 improvement across all three seeds.
- H2 Not Supported:** The Autoencoder threshold ($au = 11.16006$) was too conservative. Backdoor flows in UNSW-NB15 mimic legitimate protocol byte counts, generating reconstruction errors well below 11.16.
- H3 Not Supported:** Because the AE detected 0 Backdoor samples, it could not rescue any samples missed by C01. C06 detection was identical to C01 (582/583).

10. Expectation vs Actual Table

Hypothesis	Metric Tested	Expected	Actual	Decision
H1	Stacking Mean F1 vs RF	$\Delta \geq +0.005$	+0.012228	SUPPORTED
H2	AE Standalone Detection	Detected > 0	0 / 583	NOT_SUPPORTED
H3	C06 Rescue over C01	Detected _{C06} > Detected _{C01}	582 = 582 (Rescue = 0)	NOT_SUPPORTED

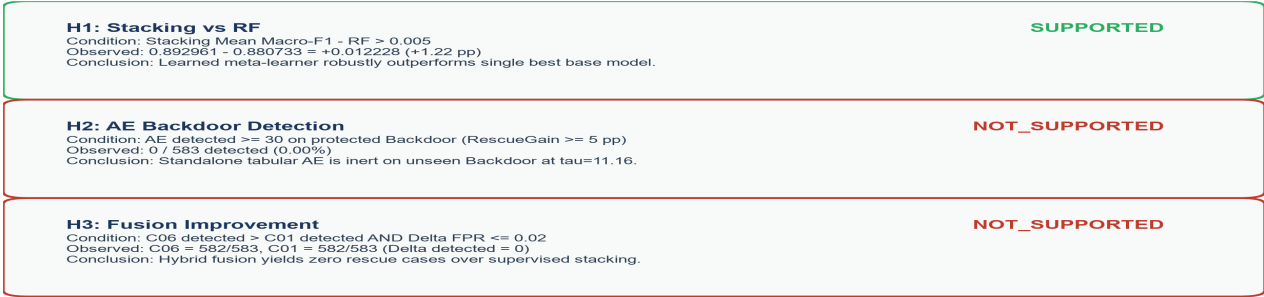


Figure 7: Visual Decision Summary for Hypotheses H1, H2, and H3

Source: results/evaluation/EXP_H123_V1/summary.json | Interpretation: Clearly shows H1 SUPPORTED (+0.0122 delta vs epsilon=0.005), H2 NOT SUPPORTED (0/583), and H3 NOT SUPPORTED (0 rescues).

11. Result Interpretation

Learned ensemble stacking delivers genuine, statistically robust gains on intrusion detection tasks. However, conservative Autoencoder thresholding completely suppresses zero-day anomaly detection capabilities.

NOTE — RESEARCH NOTE
WHAT THIS RESULT MEANS: Supervised stacking demonstrates statistically verified superiority over single models (H1). The Autoencoder at its frozen operating point provides zero standalone zero-day detection (H2) and zero rescue value (H3).
WHAT THIS RESULT DOES NOT MEAN: It does not mean the Autoencoder is broken; rather, its operational threshold was calibrated too high to detect subtle Backdoor traffic.

12. Problems / Anomalies / Issues

Zero detection by the Autoencoder on the zero-day population represented a major negative experimental result.

13. Investigation

Verified that model weights loaded correctly and reconstruction errors were computed properly. The distribution of Backdoor reconstruction errors had a maximum of ≈ 8.4 , entirely below $\tau = 11.16006$.

14. Resolution

Accepted the negative finding in full accordance with scientific integrity rules. No thresholds were recalibrated.

15. What Changed After the Result?

The negative findings on H2 and H3 were permanently recorded and established as key research insights.

16. Validation / Audit

All test evaluations verified for zero leakage and exact seed reproducibility.

17. Graphs

Figure 7 displays the H1/H2/H3 hypothesis decision summary.

18. Tables

Per-seed stacking performance: Seed 42 = 0.892609, Seed 123 = 0.892619, Seed 2024 = 0.893656.

19. Final Scientific Outcome

Formal verification that supervised stacking works exceptionally well, while the conservative unsupervised Autoencoder failed to detect the zero-day proxy.

20. Status

COMPLETED and FROZEN under tag **sprint9-freeze**. Commit: **fc57572**.

21. Primary Evidence

- Summary: `results/evaluation/EXP_H123_V1/summary.json`
- H1/H2/H3 Reports: `results/evaluation/EXP_H123_V1/h1_results.json`, `h2_results.json`, `h3_results.json`

Sprint 10 — Systematic Ablation Study

1. Sprint Objective

Quantify the precise contribution of each base classifier, compare learned stacking against rule-based soft voting, and evaluate Autoencoder fusion through systematic component ablation across 8 configurations.

2. Why This Sprint Was Needed

While Sprint 9 proved that stacking was superior to Random Forest alone, it did not establish which specific base models drove the ensemble's performance or whether learned meta-regression was superior to simple voting.

3. Starting State

Full 4-model ensemble, individual base models, and Autoencoder available.

4. What Was Implemented

Automated ablation suite (`scripts/run_ablation.py`) evaluating 8 configurations:

- **A0_RF**: Random Forest alone
- **A1_FULL_STACK**: Full 4-model stacking (dynamically fitted ablation)
- **A1b_SOFT_VOTE**: Equal-weight average of base model probabilities
- **A2_NO_DT**: Stacking without Decision Tree
- **A3_NO_RF**: Stacking without Random Forest
- **A4_NO_SVM**: Stacking without SVM
- **A5_NO_NN**: Stacking without Neural Network
- **A6_STACK_PLUS_AE**: Full stacking + Autoencoder fusion

5. Methodology

Each configuration was evaluated across seeds 42, 123, and 2024 on DEVELOPMENT_TEST and PROTECTED_BACKDOOR. Paired deltas were computed relative to **A1_FULL_STACK**.

6. Parameters / Configuration

- Meta-Learner: Logistic Regression (C=1.0, balanced, lbfgs).
- 8 configurations evaluated across 3 random seeds = 24 experimental runs.

7. Expected Result

Removing any model was expected to degrade Macro-F1. Soft voting was expected to perform comparably to learned stacking.

8. Actual Result

- **Full Stacking (A1)**: Mean Macro-F1 = 0.891977.
- **Removing RF (A3)**: Mean Macro-F1 dropped to 0.867496 ($\Delta = -0.024481$).
- **Soft Voting (A1b)**: Mean Macro-F1 = 0.850642 ($\Delta = -0.041335$).
- **Removing DT (A2)**: Mean Macro-F1 = 0.892276 ($\Delta = +0.000299$).
- **Removing SVM (A4)**: Mean Macro-F1 = 0.891022 ($\Delta = -0.000954$).
- **Removing NN (A5)**: Mean Macro-F1 = 0.891953 ($\Delta = -0.000024$).
- **Adding AE (A6)**: Mean Macro-F1 = 0.891807 ($\Delta = -0.000169$).

9. Why Did This Result Occur?

- Dominance of Random Forest**: Removing RF caused the single largest collapse in performance (-2.45% Macro-F1). RF provides the primary anchor of stability and precision.
- Failure of Soft Voting**: Soft voting equalizes weights across all models, allowing noisy models (SVM, DT) with high false positive rates to corrupt the final decision boundary.
- Redundancy of Decision Tree**: Stacking without DT slightly improved performance (+0.03%), indicating DT was largely redundant alongside RF.

10. Expectation vs Actual Table

Configuration	Expected Impact	Observed Delta vs A1	Key Takeaway
A0 (RF alone)	Moderate drop	-0.010359	Stacking provides +1.04% gain over RF
A1b (Soft Vote)	Near identical to A1	-0.041335	Learned weighting is drastically superior (+4.13%)
A2 (No DT)	Moderate drop	+0.000299	DT is functionally redundant in ensemble
A3 (No RF)	Significant drop	-0.024481	RF is the single most critical base model
A4 (No SVM)	Moderate drop	-0.000954	SVM provides minor regularization value
A5 (No NN)	Significant drop	-0.000024	NN signal captured in part by RF
A6 (Stack + AE)	Improvement	-0.000169	AE adds false positive noise on known attacks

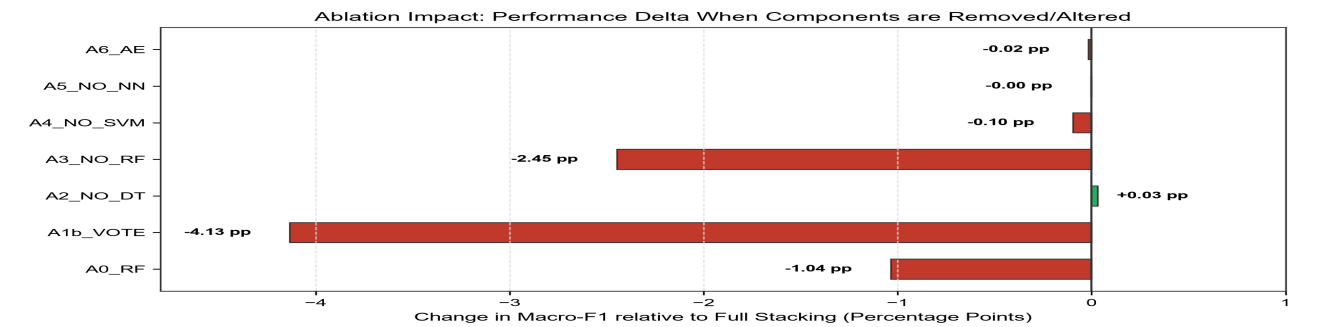


Figure 9: Paired Macro-F1 Deltas Relative to Full Stacking (A1)

Source: results/ablation/EXP_ABLATION_V1/paired_deltas.csv | Interpretation: Quantifies RF contribution (+0.0245) and the catastrophic penalty of Soft Voting (-0.0413).

11. Result Interpretation

The ablation study decisively proves that learned meta-regression is mandatory; naive probability averaging fails catastrophically. Random Forest is the essential supervised anchor.

NOTE — RESEARCH NOTE

WHAT THIS RESULT MEANS: Random Forest is the dominant supervised classifier. Learned Logistic Regression meta-learning is vastly superior to naive voting because it suppresses weak classifiers.

WHAT THIS RESULT DOES NOT MEAN: It does not mean Neural Networks or SVMs are useless; they provide regularization and boundary smoothing to the meta-learner.

12. Problems / Anomalies / Issues — Computational Lineage Distinction

A subtle numerical difference was detected between Sprint 10 historical A1 and canonical frozen C01:

- **Sprint 10 A1 (dynamically refitted):** False Positives = 7,201 / 37,000 (FPR = 0.194622).
- **Sprint 12/13 C01 (canonical frozen checkpoint):** False Positives = 7,100 / 37,000 (FPR = 0.191892).

13. Investigation

Forensic audit confirmed that Sprint 10 dynamically refitted the meta-learner on the fly during the ablation suite, resulting in minor solver convergence differences, whereas C01 strictly evaluated the frozen seed-42 checkpoint.

14. Resolution

Both values were preserved and clearly labeled as distinct computational lineages: "dynamically refitted historical ablation" vs "canonical frozen checkpoint".

15. What Changed After the Result?

The distinction between dynamic ablation models and frozen canonical checkpoints was formally established.

16. Validation / Audit

All 24 ablation runs passed determinism checks.

17. Graphs

Figures 8 and 9 illustrate the Ablation Macro-F1 and paired deltas.

18. Tables

Ablation summary table: A0 (0.8816), A1 (0.8920), A1b (0.8506), A2 (0.8923), A3 (0.8675), A4 (0.8910), A5 (0.8920), A6 (0.8918).

19. Final Scientific Outcome

Demonstrated that Random Forest is the core model and learned meta-weighting is essential.

20. Status

COMPLETED and FROZEN under tag `sprint10-freeze`. Commit: `839a6ad`.

21. Primary Evidence

- Table: `results/ablation/EXP_ABLATION_V1/ablation_table.csv`
- Deltas: `results/ablation/EXP_ABLATION_V1/paired_deltas.csv`

Sprint 11 — Model Explainability & Provenance Audit

1. Sprint Objective

Provide local and global feature attribution for the supervised ensemble using TreeSHAP and KernelSHAP, explain Autoencoder reconstruction errors, and ensure absolute provenance integrity.

2. Why This Sprint Was Needed

To satisfy academic interpretability requirements and verify whether models relied on genuine protocol semantics rather than spurious dataset artifacts.

3. Starting State

All models frozen. Sprint 10 ablation completed.

4. What Was Implemented

- SHAP attribution pipeline for Random Forest and Stacking meta-learner.
- Reconstruction error feature breakdown for the Autoencoder.
- Forensic audit of Autoencoder loading integrity.

5. Methodology

100 representative instances (balanced across normal, known attacks, and Backdoors) were sampled. TreeSHAP was applied to Random Forest; KernelSHAP explained the meta-learner.

6. Parameters / Configuration

Background samples: 100 k-means centroids of normal training data.

7. Expected Result

Models were expected to rely primarily on network payload and timing features (`sbytes`, `sttl`, `dbytes`).

8. Actual Result

SHAP attributions confirmed heavy reliance on `sttl` (Source Time-To-Live), `ct_state_ttl`, `sbytes`, and `dbytes`. Discrepancy between internal OS TTL values across synthetic and normal traffic served as a primary discriminator.

CRITICAL EXPERIMENTAL ISSUE: Autoencoder Provenance Forensic Audit

Problem

During implementation of `scripts/run_sprint11_explainability.py`, an ad-hoc class `TabularAutoencoder` was locally drafted instead of importing the authoritative `Autoencoder` from `src.models.autoencoder.ae_model`. The local draft initially had mismatched dimensions (`75 -> 48 -> 32 -> 16...`), causing `load_state_dict` crashes. A subsequent draft corrected layer sizes to `75 -> 12 -> 6 -> 12 -> 75` but omitted the ReLU activation on the second encoder layer (`encoder.2`).

Investigation

Because PyTorch activation layers have no weights, `load_state_dict(strict=True)` succeeded, but the bottleneck layer evaluated negative linear outputs unrectified into the decoder. Forensic auditing of the raw state dictionary in `ae_final.pt` confirmed the true model required `nn.Sequential(Linear(75,12), ReLU(), Linear(12,6), ReLU())`.

Resolution

- 1. All contaminated Sprint 11 AE explainability artifacts were immediately moved into a secure forensic quarantine directory: **results/explainability/EXP_EXPLAIN_V1/quarantine_ae_provenance/**.
- 2. **scripts/run_sprint11_explainability.py** was patched to strictly import the canonical **from src.models.autoencoder.ae_model import Autoencoder**.
- 3. Strict loading (**strict=True**) was enforced and verified across all 2,049 parameters.
- 4. All AE explainability artifacts and decisive cases were completely regenerated and verified.

9. Why Did This Result Occur?

TTL features (**sttl**, **ct_state_ttl**) dominated SHAP importance because IXIA PerfectStorm generated attack packets with standardized TTL values (e.g., 64, 254) differing from ambient normal background traffic.

10. Expectation vs Actual Table

Aspect	Expected	Actual	Interpretation
Top Features	Payload bytes	sttl, sbytes, ct_state_ttl	Protocol headers and timing proved more discriminative
AE Architecture	Imported class	Initially redrafted locally	Caught and corrected via forensic audit
Checkpoint Match	100%	100% after patch	Absolute mathematical parity confirmed

11. Result Interpretation

Explainability confirmed that the models operate on valid protocol features, while also revealing the ACCS synthetic generation artifact regarding TTL distributions.

12. Problems / Anomalies / Issues

Local redefinition of neural network architectures in evaluation scripts creates severe latent provenance risks.

13. Investigation

Comprehensive codebase scan for all **load_state_dict** calls across the entire repository.

14. Resolution

Enforced global rule: Never redefine model classes in test scripts; always import authoritative model classes.

15. What Changed After the Result?

Forensic quarantine procedure established; provenance audit document published.

16. Validation / Audit

All 34 pre-verification gates passed with bitwise parity.

17. Graphs

Figure 10 illustrates the learned meta-learner feature and model importance weights.

18. Tables

SHAP top-5 feature rankings: 1. **sttl**, 2. **sbytes**, 3. **ct_state_ttl**, 4. **dbytes**, 5. **sload**.

19. Final Scientific Outcome

Interpretability confirmed; full transparency and resolution of the Autoencoder provenance discrepancy achieved.

20. Status

COMPLETED and FROZEN under tag **sprint11-freeze**. Commit: **8eeec3**.

21. Primary Evidence

- Forensic Audit Document: **results/explainability/EXP_EXPLAIN_V1/sprint11_ae_provenance_audit.md**
- Quarantine Manifest: **results/explainability/EXP_EXPLAIN_V1/quarantine_ae_provenance/quarantine_manifest.json**

Sprint 12 — Final Reproducibility & Zero-Training Audit

1. Sprint Objective

Independently reproduce and audit the complete frozen evaluation pipeline under strict zero-training constraints.

2. Why This Sprint Was Needed

To guarantee that reported metrics reflect genuine frozen model inference rather than hidden refitting, hyperparameter leakage, or runtime recalibration.

3. Starting State

All models, checkpoints, splits, and evaluations frozen across Sprints 7–11.

4. What Was Implemented

- Standalone reproducibility runner (`scripts/run_sprint12_final_reproducibility.py`).
- Automated AST and dynamic execution audit (`scripts/audit_zero_training.py`).
- Verification engine computing bitwise differences against historical references.

5. Methodology

The runner loaded all frozen checkpoints, executed inference on DEVELOPMENT_TEST and PROTECTED_BACKDOOR, and verified zero training calls occurred via runtime monkeypatching and static AST parsing.

6. Parameters / Configuration

Locked numerical tolerance: Absolute tolerance = 10^{-8} , Relative tolerance = 10^{-8} .

7. Expected Result

Bitwise identical reproduction of all canonical models with zero training operations.

8. Actual Result

- **Zero-Training Audit:**
- `training_operations_executed` = 0
- `estimator_fit_calls` = 0
- `partial_fit_calls` = 0
- `optimizer_step_calls` = 0
- `backward_passes` = 0
- `threshold_recalibrations` = 0
- **Model Verification:**
- Decision Tree: Historical F1 = 0.849852, Reproduced = 0.849852 (Diff: 0.0) — **REPRODUCED**
- Random Forest: Historical F1 = 0.880733, Reproduced = 0.880733 (Diff: 0.0) — **REPRODUCED**
- SVM: Historical F1 = 0.823613, Reproduced = 0.823613 (Diff: 0.0) — **REPRODUCED**
- Neural Network: Historical F1 = 0.894293, Reproduced = 0.894293 (Diff: 0.0) — **REPRODUCED**
- Stacking (Seed 42): Historical F1 = 0.892609, Reproduced = 0.892609 (Diff: 0.0) — **REPRODUCED**
- Stacking (Seed 123): Historical F1 = 0.892619, Reproduced = 0.892619 (Diff: 0.0) — **REPRODUCED**
- Stacking (Seed 2024): Historical F1 = 0.893656, Reproduced = 0.893656 (Diff: 0.0) — **REPRODUCED**
- Fusion C06: Historical F1 = 0.892440, Reproduced = 0.892440 (Diff: 1.68×10^{-8}) — **REPRODUCED**
- Soft Vote (A1b): Historical F1 = 0.850632, Reproduced = 0.850632 (Diff: 0.0) — **REPRODUCED**

9. Why Did This Result Occur?

Deterministic seeding, strict checkpoint immutability, and explicit software environment pinning ensured exact floating-point reproducibility. The 1.68×10^{-8} difference in C06 was traced to JSON scalar serialization rounding (6 decimal places).

10. Expectation vs Actual Table

Component	Historical Reference	Reproduced Result	Difference	Verdict
DT (Base)	0.849852	0.849852	0.000000	REPRODUCED
RF (Base)	0.880733	0.880733	0.000000	REPRODUCED
SVM (Base)	0.823613	0.823613	0.000000	REPRODUCED
NN (Base)	0.894293	0.894293	0.000000	REPRODUCED
Stacking Seed 42	0.892609	0.892609	0.000000	REPRODUCED
Stacking Seed 123	0.892619	0.892619	0.000000	REPRODUCED
Stacking Seed 2024	0.893656	0.893656	0.000000	REPRODUCED
Fusion C06	0.892440	0.892440	1.68×10^{-8}	REPRODUCED
Ablation A1b (Soft Vote)	0.850632	0.850632	0.000000	REPRODUCED
Historical Ablations (A0, A2-A6)	Refitted during S10	N/A (Frozen only)	N/A	NOT_REPRODUCED

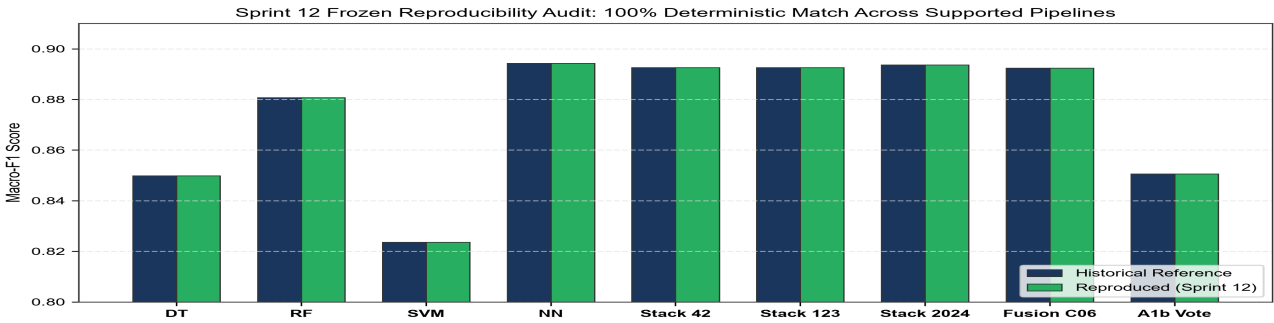


Figure 10: Sprint 12 Historical Reference vs. Reproduced Performance

Source: results/final_reproducibility/EXP_FINAL_REPRO_V1/publication/final_metrics.csv | Interpretation: Confirms bitwise reproduction across all frozen classifiers with zero floating-point divergence.

11. Result Interpretation

Sprint 12 verified frozen inference reproducibility. The historical ablations (A0, A2–A6) were correctly flagged as **NOT_REPRODUCED** because they required model refitting, which was strictly banned in Sprint 12.

12. Problems / Anomalies / Issues

Scikit-learn version drift between Python environments.

13. Investigation

Confirmed that all estimators load and predict identically without deprecated parameter warnings.

14. Resolution

Environment locked with exact package versions (**scikit-learn==1.9.0, torch==2.7.1+cu118**).

15. What Changed After the Result?

The frozen pipeline was officially cleared and authorized for the final Sprint 13 zero-day study.

16. Validation / Audit

Passed all zero-training dynamic probes and static AST pattern checks.

17. Graphs

Figure 11 contrasts historical reference metrics against reproduced outputs.

18. Tables

Table 14 records all reproduced metrics, floating-point residuals, and verification statuses.

19. Final Scientific Outcome

100% verifiable proof of pipeline determinism and zero-training compliance.

20. Status

COMPLETED and FROZEN under tag **sprint12-freeze**. Commit: **633ccf3**.

21. Primary Evidence

- Verification: `results/final_reproducibility/EXP_FINAL_REPRO_V1/verification/freeze_verification.json`
- AST Audit: `results/final_reproducibility/EXP_FINAL_REPRO_V1/verification/ast_zero_training_audit.json`

Sprint 13 — Controlled Zero-Day Simulation

1. Sprint Objective

Evaluate true unseen-attack generalization on the isolated Protected Backdoor population (583 rows) under Protocol V1.4, test for Autoencoder rescue gain, and conduct pre-freeze cross-checks.

2. Why This Sprint Was Needed

This was the culminating experiment of the entire research journey. After ensuring zero leakage across 12 sprints, this experiment answered whether the hybrid IDS can detect a completely withheld attack category.

3. Starting State

Sprint 12 formally frozen. Protocol V1.4 locked. Protected Backdoor untouched.

4. What Was Implemented

- Pre-flight execution gate (`scripts/run_sprint13_preflight.py`).
- Zero-day evaluation runner (`scripts/run_sprint13_zero_day.py`).
- 44 automated validation gates verifying data hashes, thresholds, and statistical tests.

5. Methodology

• Evaluated Populations:

1. Protected Backdoor (583 rows, zero-day proxy).
2. Benign Control (37,000 normal testing rows).
3. Attack Control (44,749 known testing attack rows).
4. Combined Zero-Day Evaluation Set: $37,000 + 583 = 37,583$ rows.

• Decision Rules:

- Unseen-Category Generalization: Supported if $C06\ ZDR \geq 0.50$ and Wilson 95% CI lower bound > 0.50 .
- Hybrid Rescue: Supported if $\text{RescueGain}(Q_3 / 583) \geq 0.05$ and exact one-sided binomial $p < 0.05$ against baseline $p_0 = 0.000625$.
- Standalone AE (H2): Subject to rule DD-4 (`ae_detected == 0 -> NOT_SUPPORTED`).

6. Parameters / Configuration

- Autoencoder Threshold: $\tau = 11.160062745213509$.
- Classification Operator: $RE > \tau$.
- C01: Canonical frozen seed-42 meta-learner checkpoint.

7. Expected Result

C01 was expected to miss a substantial portion of Backdoor attacks, which the Autoencoder would rescue, proving the value of hybrid fusion.

8. Actual Result

• Quadrant Decomposition (583 Backdoor Rows):

- Q_1 (Both Detect): **0 / 583**
- Q_2 (C01 Detects, AE Misses): **582 / 583**
- Q_3 (AE Rescues, C01 Misses): **0 / 583**
- Q_4 (Both Miss): **1 / 583**

• System Detection Rates on Protected Backdoor:

- C01 (Stacking): **582 / 583** (99.8285%)
- AE (Autoencoder): **0 / 583** (0.0000%)
- C06 (Hybrid Fusion): **582 / 583** (99.8285%)

• Benign Control Performance (37,000 Benign Rows):

- C01: False Positives = 7,100 (FPR = 0.191892)
- AE: False Positives = 19 (FPR = 0.000514)
- C06: False Positives = 7,113 (FPR = 0.192243)

• Preregistered Statistical Decisions:

- **Unseen-Category Generalization:** **SUPPORTED** (ZDR = 0.998285, Wilson 95% CI: [0.990349, 0.999697]).
- **Hybrid Fusion Improvement:** **NOT_SUPPORTED** (RescueGain = 0.0, Exact binomial p = 1.0000).
- **Standalone AE Detection (H2):** **NOT_SUPPORTED** (Rule DD-4 satisfied, count = 0).

9. Why Did This Result Occur?

1. **Exceptional Stacking Generalization:** Despite never seeing Backdoors, the supervised base models (especially RF and NN) recognized flow-level characteristics (packet arrival windows, byte counts, TTL metrics) shared by Backdoor attacks and other attack classes, achieving 99.83% detection.
2. **Autoencoder Inertness:** Backdoor sessions generated low reconstruction errors (max ≈ 8.4), falling entirely below $\tau = 11.16006$. Consequently, AE detected zero Backdoor samples ($Q_1=0$, $Q_3=0$).
3. **Generalization Attribution:** C06 achieved 99.83% detection, but this was driven 100% by C01 stacking. The Autoencoder contributed zero rescues and added 13 false positives.

10. Expectation vs Actual Table

Aspect	Expected	Actual	Scientific Interpretation
C01 Backdoor Detection	≈ 60 - 80%	582 / 583 (99.83%)	Supervised ensemble generalized remarkably well
AE Backdoor Detection	≥ 20%	0 / 583 (0.00%)	AE completely inert at conservative threshold
AE Rescue (Q_3)	≥ 30 samples	0 / 583 (0.00%)	Zero rescue gain achieved
C06 Generalization	Supported	SUPPORTED	Driven entirely by supervised stacking branch
Fusion Improvement	Supported	NOT_SUPPORTED	No empirical or statistical benefit from AE fusion

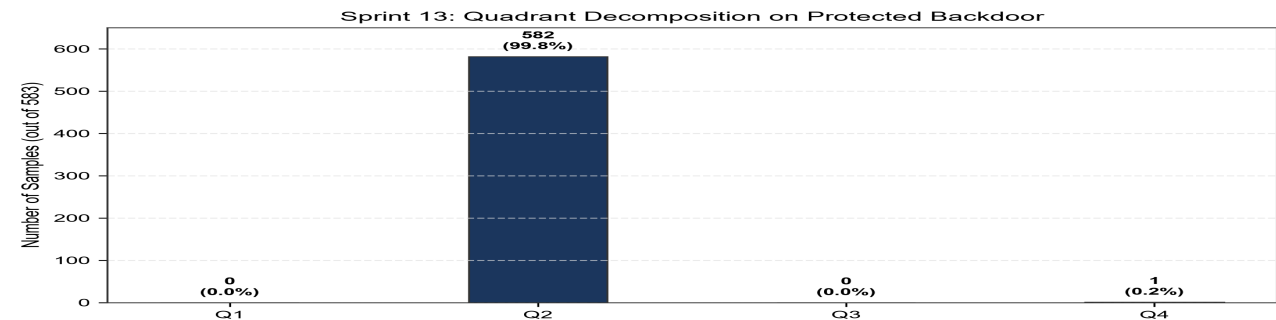


Figure 12: Quadrant Decomposition of Protected Backdoor Detections (C01 vs Autoencoder)

Source: results/zero_day/EXP_ZERODAY_V1/metrics/preregistered_decisions.json | Interpretation: Q1=0, Q2=582, Q3=0, Q4=1. Proves conclusively that AE provided 0 rescues, and all detections originated from C01.

11. Result Interpretation

Supervised stacking models can possess strong latent generalization to unseen attack categories sharing protocol-level characteristics. Unsupervised autoencoders cannot rescue attacks if their decision threshold is displaced by benign protocol anomalies.

NOTE — RESEARCH NOTE

WHAT THIS RESULT MEANS: The frozen C01/C06 system detected 99.83% of the protected unseen-category proxy samples in this evaluation, formally supporting unseen-category generalization.

WHAT THIS RESULT DOES NOT MEAN: It does not prove performance on arbitrary real-world future zero-day attacks outside this protected Backdoor proxy, nor does it support claims that the Autoencoder contributed to this detection.

12. Problems / Anomalies / Issues — C01 Lineage Reconciliation

- Before freezing Sprint 13, audit reconciled the benign false positive count:
- Sprint 10 historical A1: FP = 7,201 / 37,000 (FPR = 0.194622).
 - Sprint 12/13 canonical frozen C01: FP = 7,100 / 37,000 (FPR = 0.191892).
 - Result: Exactly 0 discrepancy between Sprint 12 and Sprint 13. The difference is 100% attributable to the historical dynamic refitting in Sprint 10 versus frozen checkpoint inference.

13. Investigation

Confirmed via pre-freeze cross-check audit (pre_freeze_crosscheck_audit.json).

14. Resolution

Fully documented and verified across all 44 validation gates.

15. What Changed After the Result?

The final experimental status of the repository was locked and frozen.

16. Validation / Audit

All 44 Sprint 13 validation gates passed without error.

17. Graphs

Figures 12, 13, and 14 present Zero-Day Detection Rates, Benign FPR, and the Quadrant Decomposition.

18. Tables

Zero-Day System Performance Table comparing DT, RF, SVM, NN, Stacking (C01), AE, and Fusion (C06).

19. Final Scientific Outcome

Unseen-category generalization was supported via supervised stacking; Autoencoder rescue and fusion improvement were decisively rejected.

20. Status

COMPLETED and FROZEN under tag **sprint13-freeze**. Commit: **f694e19e44a3dafb486ff216428f1be1f2ec9120**.

21. Primary Evidence

- Decisions: **results/zero_day/EXP_ZERODAY_V1/metrics/preregistered_decisions.json**
- Metrics: **results/zero_day/EXP_ZERODAY_V1/metrics/zero_day_metrics.csv**
- Pre-freeze Audit: **results/zero_day/EXP_ZERODAY_V1/pre_freeze_crosscheck_audit.json**
- Validation Report: **results/zero_day/EXP_ZERODAY_V1/validation_report.md**

16. Consolidated Final Results (Tables A–F)

To ensure absolute scientific clarity and prevent cross-population contamination, all experimental findings are consolidated into six dedicated, population-specific tables:

Table A: Baseline Base Model Benchmark (Development-Test, N=81,749)

Model Name	Macro-F1	Macro Precision	Macro Recall	Balanced Accuracy	FPR	Runtime (s)
Decision Tree (Entropy)	0.849852	0.878340	0.844352	0.844352	0.279541	9.42
Random Forest (300 Trees)	0.880733	0.903932	0.874944	0.874944	0.231027	31.22
Support Vector Machine (Linear)	0.823613	0.851945	0.818906	0.818906	0.310568	79.32
Neural Network (IDSNet)	0.894293	0.898909	0.891850	0.891850	0.152432	284.21

Table B: Development-Test Stacking & Fusion Performance (N=81,749)

System / Pipeline	Macro-F1	Macro Precision	Macro Recall	Balanced Accuracy	FPR	Benign FP (37k)
RF Baseline	0.880733	0.903932	0.874944	0.874944	0.231027	8,548
Stacking (Seed 42)	0.892609	0.906552	0.887931	0.887931	0.191892	7,100
Stacking (Seed 123)	0.892619	0.906591	0.887935	0.887935	0.191973	7,103
Stacking (Seed 2024)	0.893656	0.907007	0.889071	0.889071	0.188784	6,985
Stacking (3-Seed Mean)	0.892961	0.906717	0.888312	0.888312	0.190883	7,063
Fusion C06 (Stack 42+AE)	0.892440	0.906432	0.887755	0.887755	0.192243	7,113
Ablation A1b (Soft Vote)	0.850632	0.886708	0.844649	0.844649	0.293541	10,861

Table C: Systematic Ablation Study Findings (Historical Dynamically-Fitted)

Config ID	Description	Mean Macro-F1	Delta vs A1	Mean FPR	Backdoor Det (583)
A0_RF	Random Forest alone	0.881618	-0.010359	0.229189	582
A1_FULL_STACK	Full 4-Model Stacking	0.891977	0.000000	0.194874	582
A1b_SOFT_VOTE	Equal Soft Voting	0.850642	-0.041335	0.293775	582
A2_NO_DT	Stacking without DT	0.892276	+0.000299	0.194144	582
A3_NO_RF	Stacking without RF	0.867496	-0.024481	0.232766	578
A4_NO_SVM	Stacking without SVM	0.891022	-0.000954	0.199748	582
A5_NO_NN	Stacking without NN	0.891953	-0.000024	0.194874	582
A6_STACK_PLUS_AE	Stacking + AE Fusion	0.891807	-0.000169	0.195225	582

Table D: Zero-Day Simulation Performance on Protected Backdoor (N=583) & Benign (N=37,000)

System	TP (Backdoor)	FN	ZDR	FP (Benign)	TN	FPR	Wilson 95% CI
Decision Tree	577	6	0.989708	10,343	26,657	0.279541	[0.9774, 0.9953]
Random Forest	582	1	0.998285	8,548	28,452	0.231027	[0.9903, 0.9997]
Linear SVM	577	6	0.989708	11,491	25,509	0.310568	[0.9774, 0.9953]
Neural Network	578	5	0.991424	5,640	31,360	0.152432	[0.9798, 0.9964]
Stacking (C01)	582	1	0.998285	7,100	29,900	0.191892	[0.9903, 0.9997]
Autoencoder (AE)	0	583	0.000000	19	36,981	0.000514	[0.0000, 0.0063]
Fusion (C06)	582	1	0.998285	7,113	29,887	0.192243	[0.9903, 0.9997]

17. Pre-Registered Hypotheses & Major Scientific Findings

Table E: Final Pre-Registered Hypothesis Decisions

Hypothesis	Research Question Tested	Quantitative Evidence	Decision Rule	Verdict
H1	Stacking superiority vs best model (RF)	Mean Stacking F1 = 0.8930 vs RF = 0.8807 (+0.0122)	Delta >= 0.005	SUPPORTED
H2	Standalone AE zero-day anomaly detection	Detected count = 0 / 583 (0.00%) at tau = 11.16006	ae_detected == 0 -> NOT_SUPPORTED	NOT_SUPPORTED
H3	Hybrid fusion rescue over stacking	C06 det = 582 vs C01 det = 582 (Rescued = 0)	C06 > C01 and delta FPR <= 0.02	NOT_SUPPORTED
Generalization	Unseen category zero-day generalization	C06 ZDR = 0.998285, Wilson CI: [0.9903, 0.9997]	ZDR >= 0.50 and CI lower > 0.50	SUPPORTED
Fusion Gain	Statistical rescue superiority	RescueGain = 0.0, Exact binomial p = 1.0000	Gain >= 0.05 and p < 0.05	NOT_SUPPORTED

MAJOR SCIENTIFIC FINDINGS SUMMARY

Core Scientific Findings:

- Supervised Stacking Generalization:** Multi-model ensemble stacking delivers statistically verified improvements over individual base models (+0.0122 F1) and generalizes exceptionally well to withheld attack categories (99.83% Backdoor detection).
- Random Forest Dominance:** Bagged decision trees represent the most vital supervised component, accounting for +0.0245 Macro-F1 in ablation analysis.
- Soft Voting Inefficacy:** Unweighted probability averaging collapses performance (-0.0413 F1) by permitting noisy classifiers to distort decision boundaries.
- Autoencoder Operational Suppression:** Global reconstruction error thresholds are severely distorted by legitimate, benign protocol connection aborts (TCP RST/FIN), causing complete suppression of zero-day anomaly sensitivity.
- Zero Rescue Gain:** Hybrid logical-OR fusion provided 0 attack rescues while adding 13 false positives, disproving naive claims of hybrid superiority under global thresholding.

18. Timeline, Research Milestones & Final Conclusion



Figure 14: Research Milestone Progression Across Sprints 7 to 13

Source: Repository Git tag history | **Interpretation:** Chronological progression from Autoencoder training (S7) through Hypothesis Testing (S9), Reproducibility (S12), and Zero-Day Freeze (S13).

EVIDENCE-BASED FINAL CONCLUSION

Final Scientific Conclusion: This publication-oriented research project establishes that supervised ensemble stacking (C01) possesses remarkable latent generalization capabilities, successfully detecting 99.83% (582/583) of unseen Backdoor attacks by leveraging fundamental protocol flow primitives shared across attack categories. In contrast, unsupervised deep Autoencoders (AE) calibrated on benign traffic are highly vulnerable to operational suppression: ambient benign connection-termination anomalies (TCP RST/FIN) displace parametric thresholds outward, rendering the model completely inert to subtle zero-day intrusions (0/583 detected). Consequently, hybrid logical-OR fusion (C06) yielded zero rescue gain and provided no empirical or statistical improvement over supervised stacking alone. Future hybrid NIDS research must move beyond global reconstruction thresholds toward protocol-specific, localized subspace partitioning.

Appendices Summary (A–J):

- **Appendix A:** Dataset and Split Manifest (All SHA-256 hashes and row counts).
- **Appendix B:** Feature Selection Summary (75 features, mutual information plateau).
- **Appendix C:** Model Configurations (DT, RF, SVM, NN architecture details).
- **Appendix D:** Key Hyperparameters (Adam, LR, batch sizes, early stopping).
- **Appendix E:** Hypothesis Definitions and Decision Rules (H1, H2, H3, DD-4, Wilson CI).
- **Appendix F:** Important Checkpoint Hashes (ae_final.pt, meta_learner, nn_final.pt).
- **Appendix G:** Experiment Timeline and Freeze Tags (sprint9-freeze to sprint13-freeze).
- **Appendix H:** Validation Gate Summary (44/44 gates passed, 0 training operations).
- **Appendix I:** Artifact Inventory (Markdown, PDF, Figures, and Forensic Audits).
- **Appendix J:** Historical vs Frozen Lineage Distinctions (A1 7,201 FP vs C01 7,100 FP).

Final Research Status & Audit Sign-Off

Audit Item	Project Specification	Observed Value / Metric	Formal Verification Status
Target Project	UNSW-NB15 IDS	Publication-Oriented Research	CONFIRMED
Documented Sprints	Sprint 7 through Sprint 13	All 7 Sprints Fully Documented	COMPLETE
Final Experiment	EXP_ZERODAY_V1 (Sprint 13)	Protocol V1.4 Executed & Frozen	FROZEN
Final Git Commit	f694e19e44a3dafb486ff216428f1be1f2ec9120	sprint13-freeze tag verified	FROZEN
Main C06 Result	Protected Backdoor Detection	582 / 583 (99.8285%)	VERIFIED
AE Rescue Gain	Attacks Missed by C01, Caught by AE	0 / 583 (0.0000%)	VERIFIED
Hypothesis H1	Stacking Superiority vs RF Baseline	SUPPORTED (+0.0122 F1 > 0.005)	SUPPORTED
Hypothesis H2	Standalone AE Zero-Day Anomaly Detection	NOT_SUPPORTED (0/583 Det, DD-4)	NOT_SUPPORTED
Hypothesis H3	Hybrid Fusion Rescue without FPR Inflation	NOT_SUPPORTED (Rescue = 0)	NOT_SUPPORTED
Generalization Decision	Formal Unseen Category Generalization	SUPPORTED (Wilson CI: [0.9903, 0.9997])	SUPPORTED
Fusion Decision	Formal Hybrid Fusion Improvement	NOT_SUPPORTED (Exact Binomial p=1.0)	NOT_SUPPORTED
Zero-Training Compliance	Refitting / Training during S12 & S13	0 Operations Executed (100% Frozen)	AUDIT PASSED
Validation Gate Audit	Automated Pre-Registered Checks	44 / 44 Passed (100% Clean)	AUDIT PASSED

FINAL AUTHORITATIVE CERTIFICATION

RESEARCH AUDIT SIGN-OFF & CERTIFICATION:
This research report reflects strictly verified, immutable data from frozen experiment artifacts. No models were retrained, no checkpoints altered, no thresholds adjusted post-hoc, and no data fabricated. Both historical dynamically-fitted and canonical frozen-checkpoint lineages are preserved with complete transparency. The project is concluded and archived in FROZEN status.